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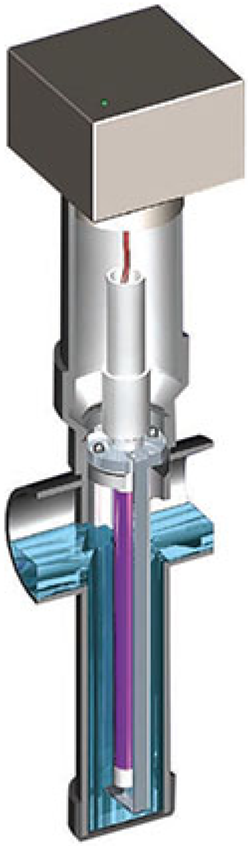
Detection of Covid-19 at Sewage Sites in Netherlands and Virus Inactivation in Wastewater through Salcor 3G UV Disinfection

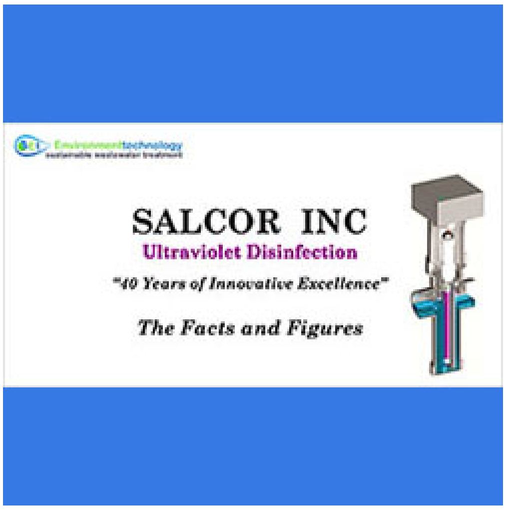


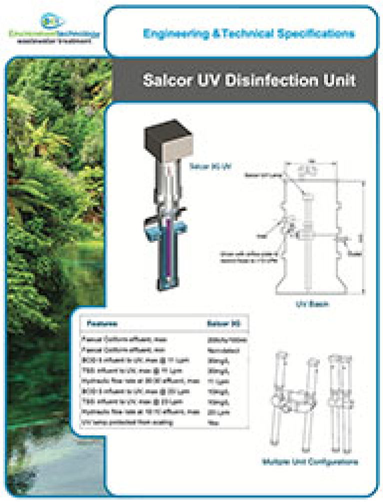
In the current COVID-19 pandemic, a significant proportion of cases shed the virus with their faeces. The first report of detection of COVID-19 in sewage came from samples taken at sewage sites in the Netherlands during February and March 2020. Sources:
Presence of Coronavirus in Sewerage and
Covid-19 Coronavirus Ultraviolet Susceptibility

COVID-19 belongs to the Coronaviridae family. The most resistant virus in the Coronaviridae family requires a 24.1 mJ/cm^2 dose of UV radiation at a wavelength of 254nm to produce a 90% reduction of activity. The Salcor 3G UV unit supplies a UV dose of >55 mJ/cm^2, meaning it is capable of producing an >90% reduction of activity.

Ultraviolet disinfection is a cost-effective way to increase wastewater treatment from Secondary to Tertiary standards through the inactivation of viruses such as COVID-19 and bacteria.

Unlike treatment by chlorination, the UV irradiation process does not produce harmful by-products meaning that the end product can be reused for sub-surface irrigation in gardens or toilet flushing. Ultraviolet light is able to inactivate microorganisms because it damages the DNA or RNA of the organism. The organism is still able to function; however, it cannot use its DNA/RNA to reproduce and so the population of the organism dies out.





[Download Information on Wastewater Treatment with UV disinfection]

[Download Engineering and Technical Specifications]

BENEFITS AND FEATURES OF UV DISINFECTION

Protects Health and Environment

- Inactivates viruses and bacteria
- Produces no harmful by-products
- Enables water-reuse, e.g. toilet-flushing and irrigation
- Alarmed for reliable and continuous performance monitoring

Residential, Commercial and Municipal Uses

- 2-year limited warranty on Salcor Model 3G UV Disinfection Unit and the UV lamp
- Units can be connected in series and/or parallel to achieve a higher flow capacity or for further risk minimisation
- Minimal footprint
- Quick installation into ground/pump chambers
- 'Long-life' UV lamp, easily replaced
- Teflon® barrier resists fouling, minimal maintenance of the UV light source
- Energy efficient, less than 40-watts or 0.96kwhrs/day

Third Party Testing and Approved

- Floodproof UL Certified Unit (NEMA 6P)
- Outstanding results when tested for 6 months with 19 different treatment systems to NSF Standard 40

Design Parameters

Max flow rate at 10:10 effluent:	23 Lpm
Max flow rate at 30:30 effluent:	11 Lpm
Wave length:	UV lamp is low pressure mercury, 90 % of output at 254 nm
Lamp power:	40 watts, 0.96 kWhr/ day assuming continuous

	operation
UV Dosage:	>55 mJ/cm^2 @1m at end of life
Lamp warranty:	2 yrs
Input voltage:	240 v at 50 Hz at up to 0.5 Amps
Contact Time:	min. 2.45 seconds at max. flow rate
Inlet and outlet pipe:	100mm Schedule 40 ABS
Ballast output frequency:	90 % efficient, high frequency operation (50 kHz) with thermal link protection
Faecal coliform reduction at lamp end-of-life (2 years):	>99.9 %

Salcor 3G UV Unit available from Environment Technology - Current price: $1,720 excl gst

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For more information about AES wastewater treatment systems visit our website or call us on 0800 927 834 (0800 waste H2O).

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